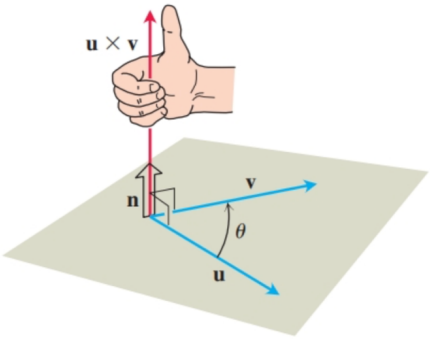
11.4 The Cross Product (The Vector Product)



Definition 1: Cross Product

$$\mathbf{u}\times\mathbf{v} = |\mathbf{u}||\mathbf{v}|(\sin\theta)\mathbf{n}\ ,\quad 0\le\theta\le\pi$$

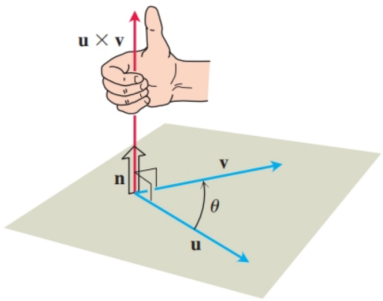
where θ is the angle between **u** and **v**, and **n** is the unit vector indicating the direction of $\mathbf{u}\times\mathbf{v}$.

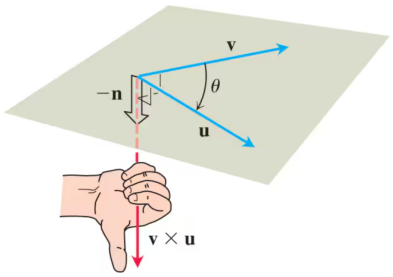
u , v and $\mathbf{u}\times\mathbf{v}$ form a right-handed system.

Remarks:

1. The cross product applies ***only*** to vectors **in space**.
2. If *u* and *v* are non parallel vectors in space, then they determine a plane.

   Vectors normal to this plane are in the direction of $\mathbf{u}\times\mathbf{v}$ or $-(\mathbf{u}\times\mathbf{v})=\mathbf{v}\times\mathbf{u}$.





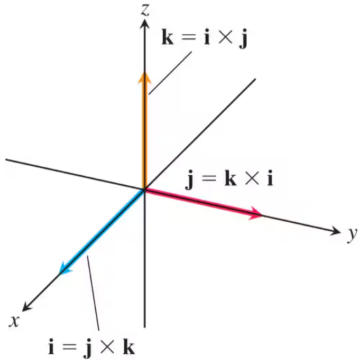
3. $(\mathbf{u}\times\mathbf{v})\perp\mathbf{u}$ and $(\mathbf{u}\times\mathbf{v})\perp\mathbf{v}$. ∴ $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{u}=0$ and $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{v}=0$.

4. (i) $\mathbf{i}\times\mathbf{j}=\mathbf{k}=-(\mathbf{j}\times\mathbf{i})$,

   $\mathbf{j}\times\mathbf{k}=\mathbf{i}=-(\mathbf{k}\times\mathbf{j})$,

   $\mathbf{k}\times\mathbf{i}=\mathbf{j}=-(\mathbf{i}\times\mathbf{k})$.

   (ii) $\mathbf{i}\times\mathbf{i}=\mathbf{j}\times\mathbf{j}=\mathbf{k}\times\mathbf{k}=\mathbf{0}$.



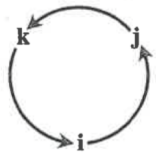
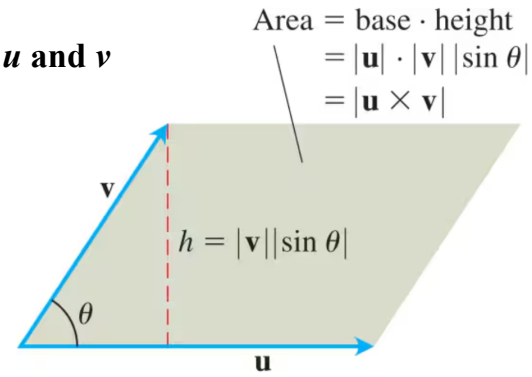


Diagram for recalling these products

5. **(i) The area of a parallelogram determined by *u* and *v* is given by** $|\mathbf{u}\times\mathbf{v}|=|\mathbf{u}||\mathbf{v}||\sin\theta|$.

   (ii) The area of a triangle determined by *u* and *v* is given by $\frac{1}{2}|\mathbf{u}\times\mathbf{v}|=\frac{1}{2}|\mathbf{u}||\mathbf{v}|\sin\theta$



6. For any nonzero vectors **u** and **v**, (i) $\mathbf{u}\ /\!/\ \mathbf{v}\ \Leftrightarrow\ \mathbf{u}\times\mathbf{v}=\mathbf{0}$.
   (Non-zero vectors **u** and **v**, are parallel if and only if $\mathbf{u}\times\mathbf{v}=\mathbf{0}$.)

   (ii) $\mathbf{u}\perp\mathbf{v}\Leftrightarrow\ \mathbf{u}\cdot\mathbf{v}=0$.

Properties of the Cross Product

If **u, v** and **w** are vectors and r, s are scalars, then

1. $(r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$
2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
3. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$
4. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = \mathbf{v} \times \mathbf{u} + \mathbf{w} \times \mathbf{u}$
5. $\mathbf{0} \times \mathbf{u} = \mathbf{0}$
6. $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$

Determinant form Formula for u × v

Let $\mathbf{u} = u_1\,\mathbf{i} + u_2\,\mathbf{j} + u_3\,\mathbf{k}$ and $\mathbf{v} = v_1\,\mathbf{i} + v_2\,\mathbf{j} + v_3\,\mathbf{k}$. Then $\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$.

Proof:

By the distributive laws and rules for multiplying **i**, **j** and **k**, we have

$$\begin{aligned}
\mathbf{u} \times \mathbf{v} &= (u_1\,\mathbf{i} + u_2\,\mathbf{j} + u_3\,\mathbf{k}) \times (v_1\,\mathbf{i} + v_2\,\mathbf{j} + v_3\,\mathbf{k}) \\
&= u_1v_1\,\mathbf{i}\times\mathbf{i} + u_1v_2\,\mathbf{i}\times\mathbf{j} + u_1v_3\,\mathbf{i}\times\mathbf{k} + u_2v_1\,\mathbf{j}\times\mathbf{i} + u_2v_2\,\mathbf{j}\times\mathbf{j} + u_2v_3\,\mathbf{j}\times\mathbf{k} \\
&\quad + u_3v_1\,\mathbf{k}\times\mathbf{i} + u_3v_2\,\mathbf{k}\times\mathbf{j} + u_3v_3\,\mathbf{k}\times\mathbf{k} \\
&= u_1v_2\,\mathbf{k} - u_1v_3\,\mathbf{j} - u_2v_1\,\mathbf{k} + u_2v_3\,\mathbf{i} + u_3v_1\,\mathbf{j} - u_3v_2\,\mathbf{i} \\
&= (u_2v_3 - u_3v_2)\,\mathbf{i} - (u_1v_3 - u_3v_1)\,\mathbf{j} + (u_1v_2 - u_2v_1)\,\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.
\end{aligned}$$

Example 1. Find $\mathbf{u}\times\mathbf{v}$ and $\mathbf{v}\times\mathbf{u}$ if $\mathbf{u} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{v} = -4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.

Solution:

$$\begin{aligned}
\mathbf{u}\times\mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix}\mathbf{k} \\
&= -2\mathbf{i} - 6\mathbf{j} + 10\mathbf{k} \\
\mathbf{v}\times\mathbf{u} &= -(\mathbf{u}\times\mathbf{v}) = 2\mathbf{i} + 6\mathbf{j} - 10\mathbf{k} \quad \textbf{Property 3}
\end{aligned}$$

Example 2

Find a vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Solution:

The vector $\overrightarrow{PQ} \times \overrightarrow{PR}$ is a vector perpendicular to the plane because it is perpendicular to both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

$$\begin{aligned}
\overrightarrow{PQ} &= (2-1)\mathbf{i} + (1+1)\mathbf{j} + (-1-0)\mathbf{k} = \mathbf{i} + 2\mathbf{j} - \mathbf{k} \\
\overrightarrow{PR} &= (-1-1)\mathbf{i} + (1+1)\mathbf{j} + (2-0)\mathbf{k} = -2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} \\
\overrightarrow{PQ} \times \overrightarrow{PR} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 2 & 2 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & -1 \\ -2 & 2 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & 2 \\ -2 & 2 \end{vmatrix}\mathbf{k} \\
&= 6\mathbf{i} + 6\mathbf{k}.
\end{aligned}$$

Example 3:
Find the area of the triangle with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Solution:
The area of the triangle with vertices P, Q, and R is

$$\frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \frac{1}{2}|6\mathbf{i} + 6k| \quad \text{(Values from Example 2)}$$
$$= \frac{1}{2}\sqrt{6^2 + 6^2} = \frac{6}{2}\sqrt{2} = 3\sqrt{2}$$

Example 4:
Find a unit vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Solution:
Since the vector $\overrightarrow{\boldsymbol{PQ}} \times \overrightarrow{\boldsymbol{PR}}$ is perpendicular to the plane, its direction $\mathbf{n}$ is a unit vector perpendicular to the plane. Taking values from Examples 2 and 3, we have

$$\mathbf{n} = \frac{\overrightarrow{PQ} \times \overrightarrow{PR}}{\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right|} = \frac{6\mathbf{i}+6\mathbf{k}}{6\sqrt{2}} = \frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{k}.$$

Triple Product (Product of 3 vectors)

Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be any three vectors in R^3.

(1) $\mathbf{u}(\mathbf{v} \cdot \mathbf{w})$ and $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ are **VECTORS**. In general, $\mathbf{u}(\mathbf{v} \cdot \mathbf{w}) \neq (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$.

Example 5:

Let $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 2\mathbf{j} + \mathbf{k}$, and $\mathbf{w} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$. Then

$\mathbf{u}(\mathbf{v} \cdot \mathbf{w}) = (\mathbf{i} + 2\mathbf{j} + \mathbf{k})(0 + 2 + 4) = 6\mathbf{i} + 12\mathbf{j} + 6\mathbf{k}$, and

$(\mathbf{u} \cdot \mathbf{v})\mathbf{w} = (0 + 4 + 1)(2\mathbf{i} + \mathbf{j} + 4\mathbf{k}) = 10\mathbf{i} + 5\mathbf{j} + 20\mathbf{k}$..

$\therefore$ $\mathbf{u}(\mathbf{v} \cdot \mathbf{w}) \neq (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ for $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 2\mathbf{j} + \mathbf{k}$, and $\mathbf{w} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$.

(2) Triple Scalar (Box) Product

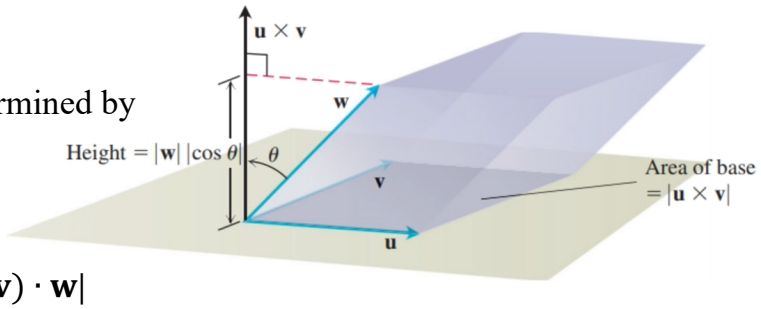
$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ **is a SCALAR.**

Notation: $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = [\mathbf{u}, \mathbf{v}, \mathbf{w}]$

Some properties of the triple scalar product:

(a) Geometric Interpretation

The volume of a parallelepiped determined by $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ is given by $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$



Volume = (area of base)(height)

$= |\mathbf{u} \times \mathbf{v}||\mathbf{w}|\cos\theta = |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$

Theorem 1: $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are coplanar $\Leftrightarrow$ $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = 0$.

❖ If any two vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are the same, then the triple scalar product is zero. That is $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$, $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$, $(\mathbf{u} \times \mathbf{u}) \cdot \mathbf{w} = 0$.

Example 6:

1. Find the volume of the box (parallelepiped) having $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = -2\mathbf{j} + \mathbf{k}$ and $\mathbf{w} = \mathbf{i} + 7\mathbf{j} - 4\mathbf{k}$ as its sides.

2. Show that the points $P(0, 0, 3)$, $Q(2, -1, 2)$, $R(3, 2, 1)$ and $S(1, 3, 2)$ are coplanar.

Solution:

1. The volume of the parallelepiped $= |(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$

$$= \left| \begin{vmatrix} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 1 & 7 & -4 \end{vmatrix} \right| = \left| -2\begin{vmatrix} 1 & -1 \\ 1 & -4 \end{vmatrix} - 3\begin{vmatrix} 1 & 2 \\ 1 & 7 \end{vmatrix} \right| = |-9| = 9 \text{ unit}^3.$$

2. $\overrightarrow{PQ} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$, $\overrightarrow{PR} = 3\mathbf{i} + 2\,\mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{PS} = \mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

$$\left| \left(\overrightarrow{PQ} \times \overrightarrow{PR}\right) \cdot \overrightarrow{PS} \right| = \left| \begin{vmatrix} 2 & -1 & -1 \\ 3 & 2 & -2 \\ 1 & 3 & -1 \end{vmatrix} \right| = \left| 2\begin{vmatrix} 2 & -2 \\ 3 & -1 \end{vmatrix} + \begin{vmatrix} 3 & -2 \\ 1 & -1 \end{vmatrix} - \begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix} \right| = |\,2(4) - 1 - 7\,| = 0.$$

Therefore $\overrightarrow{PQ}$, $\overrightarrow{PR}$ and $\overrightarrow{PS}$ lie on the same plane. Hence, P, Q, R and S are coplanar.

(b) Determinant form Formula for $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$

Let $\mathbf{u} = u_1\,\mathbf{i} + u_2\,\mathbf{j} + u_3\,\mathbf{k}$ and $\mathbf{v} = v_1\,\mathbf{i} + v_2\,\mathbf{j} + v_3\,\mathbf{k}$ and $\mathbf{w} = w_1\,\mathbf{i} + w_2\,\mathbf{j} + w_3\,\mathbf{k}$.

Then $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$.

Proof:

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \cdot (w_1\,\mathbf{i} + w_2\,\mathbf{j} + w_3\,\mathbf{k})$$

$$= \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \right) \cdot (w_1\,\mathbf{i} + w_2\,\mathbf{j} + w_3\,\mathbf{k})$$

$$= w_1 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} \cdot \mathbf{i} + w_2 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} \cdot \mathbf{j} + w_3 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} \cdot \mathbf{k}$$

$$- w_1 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} \cdot \mathbf{i} - w_2 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} \cdot \mathbf{j} - w_3 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} \cdot \mathbf{k}$$

$$+ w_1 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \cdot \mathbf{i} + w_2 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \cdot \mathbf{j} + w_3 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \cdot \mathbf{k}$$

$$= w_1 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - w_2 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + w_3 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

$$= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$

(c) Cyclic property $\boxed{(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=(\mathbf{v}\times\mathbf{w})\cdot\mathbf{u}=(\mathbf{w}\times\mathbf{u})\cdot\mathbf{v}}$

Notation: $[\mathbf{u},\mathbf{v},\mathbf{w}] = [\mathbf{v},\mathbf{w},\mathbf{u}] = [\mathbf{w},\mathbf{u},\mathbf{v}]$.

$$(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=\begin{vmatrix}u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\end{vmatrix}=-\begin{vmatrix}v_1 & v_2 & v_3\\ u_1 & u_2 & u_3\\ w_1 & w_2 & w_3\end{vmatrix}$$

$$=-\left(-\begin{vmatrix}v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\\ u_1 & u_2 & u_3\end{vmatrix}\right)=\begin{vmatrix}v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\\ u_1 & u_2 & u_3\end{vmatrix}=(\mathbf{v}\times\mathbf{w})\cdot\mathbf{u}$$

(d) Exchanging " · " and " × " $\boxed{(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})}$

$$(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}=(\mathbf{v}\times\mathbf{w})\cdot\mathbf{u}=\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$$

(from (c)) (by the commutative law).

Example 7:

If $\mathbf{u}=\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}$ and $\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}\neq 0$, find $\mathbf{u}\cdot\mathbf{a}$, $\mathbf{u}\cdot\mathbf{b}$, and $\mathbf{u}\cdot\mathbf{c}$.

Solution:

$$\mathbf{u}\cdot\mathbf{a}=\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\cdot\mathbf{a}=\frac{\mathbf{b}\times\mathbf{c}\cdot\mathbf{a}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}=\frac{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}=1,$$

$$\mathbf{u}\cdot\mathbf{b}=\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\cdot\mathbf{b}=\frac{\mathbf{b}\times\mathbf{c}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}=0 \text{ and}$$

$$\mathbf{u}\cdot\mathbf{c}=\frac{\mathbf{b}\times\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}\cdot\mathbf{c}=\frac{\mathbf{b}\times\mathbf{c}\cdot\mathbf{c}}{\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}}=0.$$

(3) Triple Vector Product $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})$

(i) In general, $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})\neq(\mathbf{u}\times\mathbf{v})\times\mathbf{w}$. (The Associative law fails)

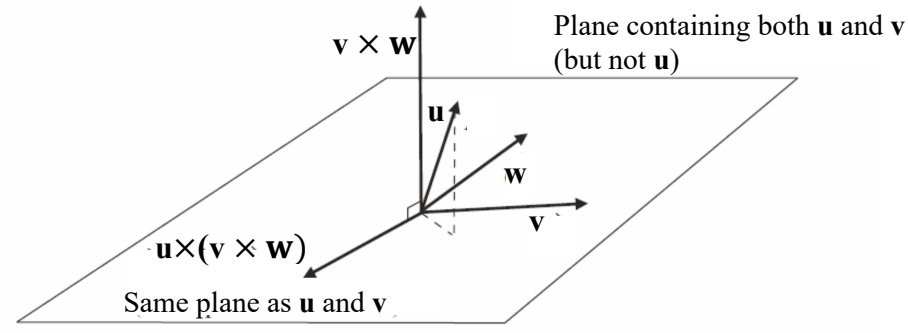
Example 8:

Let $\mathbf{u}=\mathbf{i}+\mathbf{j}$, $\mathbf{v}=\mathbf{j}+\mathbf{k}$ and $\mathbf{w}=\mathbf{i}+\mathbf{j}+\mathbf{k}$. Then

$$\mathbf{v}\times\mathbf{w}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 0 & 1 & 1\\ 1 & 1 & 1\end{vmatrix}=\mathbf{j}-\mathbf{k},\ \mathbf{u}\times(\mathbf{v}\times\mathbf{w})=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & 1 & 0\\ 0 & 1 & -1\end{vmatrix}=-\mathbf{i}+\mathbf{j}+\mathbf{k}.$$

$$\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & 1 & 0\\ 0 & 1 & 1\end{vmatrix}=\mathbf{i}-\mathbf{j}+\mathbf{k},\ (\mathbf{u}\times\mathbf{v})\times\mathbf{w}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & -1 & 1\\ 1 & 1 & 1\end{vmatrix}=-2\mathbf{i}+2\mathbf{k}\neq\mathbf{u}\times(\mathbf{v}\times\mathbf{w}).$$

(ii) $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ is in the plane S that containing the vectors $\mathbf{v}$ and $\mathbf{w}$.



(iii) $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$.

Example 9:

For any vectors **u**, **v** and **w**, show that $(\mathbf{u} \times \mathbf{v}) \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{w} \cdot \mathbf{u} \times \mathbf{v})\mathbf{v}$.

Solution:

Let $\mathbf{z} = \mathbf{u} \times \mathbf{v}$. Then $(\mathbf{u} \times \mathbf{v}) \times (\mathbf{v} \times \mathbf{w}) = \mathbf{z} \times (\mathbf{v} \times \mathbf{w})$

$$= (\mathbf{z} \cdot \mathbf{w})\mathbf{v} - (\mathbf{z} \cdot \mathbf{v})\mathbf{w}$$

$$= (\mathbf{u} \times \mathbf{v} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \times \mathbf{v} \cdot \mathbf{v})\mathbf{w}$$

$$= (\mathbf{w} \cdot \mathbf{u} \times \mathbf{v})\mathbf{v}.$$